

A woman in a white lab coat is seated at a desk, looking at a laptop. In the background, a presentation screen displays the text "Introduction to AI Ethics".

ASSET: AI for Skills, Sustainability and Training

Self-Learning Nugget Series in TVET Teacher Education

Welcome to the comprehensive course information guide for the ASSET project, a groundbreaking initiative designed to transform vocational education through artificial intelligence. This innovative programme represents a collaborative effort between UNESCO-UNEVOC Network partners to equip TVET educators with essential AI literacy skills and practical tools for the digital age.

Project Background & Global Initiative

The ASSET project emerges as a pioneering component of the UNESCO-UNEVOC Network Coaction Initiative, representing a truly international collaboration. Led by HWK OWL (Chamber of Skilled Crafts Ostwestfalen-Lippe zu Bielefeld), this ambitious undertaking unites expertise from Magdeburg, Sri Lanka, and Mauritius UNEVOC Centres.

At its core, the project develops AI-supported, open-access self-learning materials specifically tailored for vocational teachers and trainers worldwide. The course utilises LiaScript, an innovative open-source platform that enables interactive activities, supports offline use, and facilitates multilingual adaptation to serve diverse global communities.

This initiative directly supports two critical Sustainable Development Goals: SDG 4.4, which focuses on developing skills for work and employment, and SDG 8.5, which promotes decent work opportunities for all. By bridging the gap between traditional vocational education and emerging AI technologies, ASSET positions educators at the forefront of educational innovation.



Key Partnership: UNESCO-UNEVOC Network collaboration spanning four countries and multiple expertise areas.

Strategic Objectives & Learning Outcomes

AI Literacy Development

Equip teachers and trainers with comprehensive AI literacy skills, enabling them to understand, evaluate, and integrate artificial intelligence technologies effectively within their educational practices.

- Foundational AI concepts
- Practical application methods
- Critical evaluation skills

Practical Tool Mastery

Provide hands-on experience with cutting-edge AI tools specifically selected for their relevance and applicability in vocational education and training environments.

- Interactive demonstrations
- Real-world applications
- Implementation strategies

Flexible Learning Framework

Support diverse learning needs through modular, adaptable content that accommodates different schedules, technical capabilities, and institutional requirements across global contexts.

- Self-paced progression
- Offline accessibility
- Multilingual support

These objectives work synergistically to create a comprehensive learning experience that transforms how TVET educators approach technology integration, ensuring they remain relevant and effective in an increasingly digital world.

Target Audience & Professional Communities

The ASSET programme addresses the professional development needs of a diverse and influential community within the vocational education sector. Our carefully designed content serves multiple stakeholder groups, each bringing unique perspectives and requirements to the learning experience.



TVET Teachers & Trainers

Front-line educators delivering vocational programmes across various trades and industries. These professionals directly impact learner outcomes and serve as the primary bridge between traditional craft knowledge and emerging technological capabilities. Their engagement ensures practical, classroom-tested applications of AI tools.



Curriculum Designers

Educational architects responsible for structuring learning pathways and integrating new technologies into existing programmes. Their participation ensures systematic integration of AI literacy across vocational qualifications and maintains alignment with industry standards and employment requirements.



Instructional Technologists

Technical specialists who facilitate technology adoption and provide ongoing support for digital learning initiatives. Their expertise ensures effective implementation strategies and sustainable technology integration within institutional frameworks.



Education Policymakers

Decision-makers who shape educational policy and resource allocation at institutional, regional, and national levels. Their involvement ensures strategic alignment and sustainable funding for AI integration initiatives within broader educational development frameworks.

Module 1: Orientation – Setting the Foundation

The Orientation module serves as your comprehensive introduction to artificial intelligence within the vocational education and training context. This foundational unit establishes essential understanding and creates shared vocabulary amongst participants from diverse professional backgrounds.

Participants explore the fundamental rationale behind AI integration in TVET settings, examining both opportunities and challenges. The module addresses common misconceptions whilst building confidence in approaching AI technologies. Through structured activities and reflective exercises, learners develop a clear understanding of how AI can enhance rather than replace traditional teaching methodologies.

Key learning outcomes include understanding project goals and framework, mastering basic AI definitions and terminology, recognising AI's relevance to vocational training contexts, and developing realistic expectations for technology integration. The module concludes with practical orientation exercises that prepare participants for subsequent technical modules.

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📅 **Duration:** Approximately 45-60 minutes of self-paced learning with optional extended activities.

Module 2: AI Basics – Technical Foundations

The AI Basics module provides essential technical knowledge without overwhelming non-technical participants. This carefully structured unit demystifies artificial intelligence concepts through practical examples and vocational education applications.

01

Machine Learning Fundamentals

Explore how machines learn from data, with specific examples relevant to TVET contexts such as predictive maintenance in automotive training or pattern recognition in manufacturing processes.

03

Application Categories

Survey typical AI applications across various vocational sectors, from healthcare diagnostics to construction planning, demonstrating relevance across diverse TVET programmes.

Interactive exercises guide participants through hands-on exploration of AI concepts using familiar vocational examples. The module balances technical accuracy with accessibility, ensuring all participants develop sufficient understanding to engage confidently with subsequent practical modules.

02

Data Understanding

Examine the role of data in AI systems, including data types commonly found in vocational settings, quality considerations, and privacy implications for educational environments.

04

Educational Integration

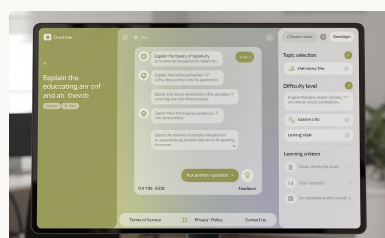
Connect technical concepts to pedagogical practice, showing how AI understanding enhances teaching effectiveness and learner engagement in vocational contexts.

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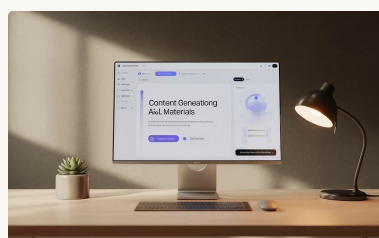
Module 3: Tool Presentations – Practical Applications

The Tool Presentations module transforms theoretical knowledge into practical capability through hands-on exploration of carefully selected AI tools. This interactive unit provides direct experience with technologies that can immediately enhance TVET teaching and learning environments.



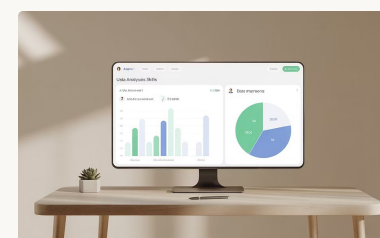
Conversational AI Platforms

Experience advanced chatbots and virtual assistants designed for educational support. Learn to leverage these tools for student queries, administrative tasks, and personalised learning assistance across various vocational disciplines.



Content Generation Tools

Explore AI-powered platforms for creating educational resources, including lesson plans, assessment questions, and multimedia content tailored to specific TVET programmes and learning outcomes.



Assessment & Feedback Systems

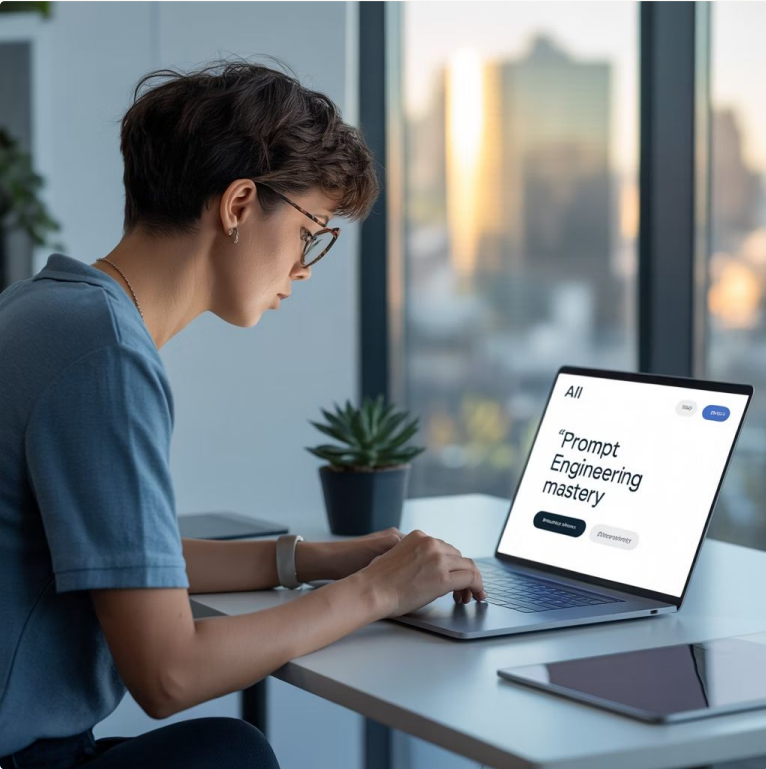
Discover intelligent assessment tools that provide automated feedback, progress tracking, and personalised learning recommendations for vocational skills development and competency measurement.

Each tool presentation includes live demonstrations, step-by-step tutorials, and practical exercises that allow participants to experiment safely within guided parameters. The module emphasises real-world application scenarios, helping educators identify which tools align best with their specific teaching contexts and institutional capabilities.

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Module 4: Prompting – Effective AI Communication



The Prompting module develops critical communication skills for effective AI interaction. This practical unit teaches participants how to construct clear, specific, and results-oriented prompts that maximise AI tool effectiveness within educational contexts.

Through structured exercises and progressive skill-building activities, participants learn to craft prompts that generate high-quality educational content, provide meaningful feedback, and support diverse learning needs. The module covers prompt anatomy, context setting, specificity techniques, and iterative refinement strategies.

✔ **Key Skill:** Master the art of prompt engineering for optimal AI interactions and educational outcomes.



Clarity & Precision

Learn to construct unambiguous prompts that clearly communicate educational objectives and desired outcomes, reducing misinterpretation and improving AI response quality.



Context Integration

Master techniques for providing relevant background information, including learner characteristics, curriculum requirements, and institutional constraints that inform AI responses.



Iterative Refinement

Develop skills for systematically improving prompts based on AI responses, creating feedback loops that enhance both immediate results and long-term interaction effectiveness.

Practical exercises include creating prompts for lesson planning, assessment design, and student support scenarios. Participants work through common prompting challenges and develop personalised strategies that align with their specific TVET contexts and teaching styles.

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Module 5: Quality & Ethics – Responsible AI Implementation

The Quality & Ethics module addresses critical considerations for responsible AI implementation in educational settings. This essential unit equips participants with frameworks for evaluating AI quality, identifying potential biases, and maintaining ethical standards throughout their professional practice.

Bias Recognition & Mitigation Understand how algorithmic bias manifests in educational AI systems and develop strategies for identifying and addressing discriminatory patterns that may disadvantage certain student groups or perpetuate existing inequalities within vocational education. - Common bias types in educational AI - Detection methodologies - Mitigation strategies - Ongoing monitoring approaches	Transparency & Accountability Explore principles of transparent AI use, including clear communication with students about AI involvement in their learning experience, maintaining human oversight in critical decisions, and establishing accountability frameworks for AI-supported educational practices. - Disclosure requirements - Human-in-the-loop protocols - Decision audit trails - Student rights protection	Quality Assurance Frameworks Develop systematic approaches for evaluating AI output quality, including accuracy verification, relevance assessment, and alignment with educational objectives. Learn to implement quality control processes that maintain educational standards whilst leveraging AI capabilities. - Output verification methods - Relevance criteria - Accuracy checking protocols - Continuous improvement cycles
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The module includes case studies from real educational implementations, ethical decision-making frameworks, and practical checklists for ongoing quality and ethics monitoring. Participants develop institutional guidelines that support responsible AI adoption within their specific contexts.

Course Features & Contact Information

100%

Online Access

Fully web-based learning platform accessible from any device with internet connectivity

24/7

Self-Paced

Learn at your own rhythm with no fixed schedule constraints or deadline pressures

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Cost Barrier

Completely free access ensuring global accessibility regardless of institutional budget

Technical Features

- **Interactive Elements:** Quizzes, exercises, and hands-on demonstrations embedded throughout all modules
- **Offline Capability:** Downloadable content for low-bandwidth or intermittent connectivity environments
- **Multilingual Support:** Adaptable framework enabling localisation for diverse global audiences
- **Open Access:** Creative Commons licensing supporting institutional customisation and local adaptation

Educational Benefits

- **Microlearning Design:** Bite-sized units that fit busy professional schedules
- **Practical Focus:** Real-world applications directly relevant to TVET contexts
- **Progressive Complexity:** Scaffolded learning that builds confidence systematically
- **Flexible Integration:** Compatible with existing professional development programmes

Project Contact

For additional information, technical support, or partnership enquiries, please contact our programme coordinator:

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